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1 Introduction

Definition 1.1. An *arithmetic function* takes $\mathbb{N}_{\geq 1}$ to $\mathbb{C}$.

Definition 1.2. An arithmetic function is called *multiplicative* if $f(mn) = f(m)f(n)$ for all coprime positive integers m and n. It is called *completely multiplicative* if this relation holds even if m and n are not coprime.

Definition 1.3. The *Dirichlet convolution* of two arithmetic functions f and g, denoted $f * g$, is defined as

$$(f * g)(n) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right),$$

where the sum is taken over positive divisors d of n.

Theorem 1.4. If two of f, g, and $f * g$ are multiplicative, so is the third.

2 First examples

Example 2.1. Compute the number of divisors of 2024 and find their sum.

Let $d(n)$ be the number of divisors of n, and let $\sigma(n)$ be the sum of the divisors of n. We seek to compute $d(2024)$ and $\sigma(2024)$. Let $\mathbf{1}$ and id be functions with $\mathbf{1}(n) = 1$ and let $\mathrm{id}(n) = n$, which are both clearly multiplicative. Now we see

$$d(n) = \sum_{d|n} 1 = \sum_{d|n} \mathbf{1}(d)\mathbf{1}\left(\frac{n}{d}\right) = (\mathbf{1} * \mathbf{1})(n).$$

Since $d(n)$ may be written as the Dirichlet convolution of two multiplicative functions, it is itself multiplicative. Similarly, observe that

$$\sigma(n) = \sum_{d|n} d = \sum_{d|n} \mathrm{id}(d)\mathbf{1}\left(\frac{n}{d}\right) = (\mathrm{id} * \mathbf{1})(n),$$

so σ is multiplicative as well.

So how do we use the multiplicativity to actually compute $d(2024)$ and $\sigma(2024)$? Well, observe that if f is a multiplicative function and n is a positive integer whose prime factorization is $p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$, we have $f(n) = f(p_1^{\alpha_1}) f(p_2^{\alpha_2}) \dots f(p_k^{\alpha_k})$. This fact is very important—if we want to compute f for arbitrary n, we only need to compute f for prime powers, and if we want to prove two multiplicative functions f and g are equal, we only need to show they are equal for prime powers. So factoring $2024 = 2^3 \cdot 11 \cdot 23$, we find

$$d(2024) = d(2^3)d(11)d(23) = (3+1)(1+1)(1+1) = 16$$

and

$$\sigma(2024) = d(2^3)d(11)d(23) = (1 + 2 + 2^2 + 2^3)(1 + 11)(1 + 23) = 4320.$$

This is another way to find the "standard" formulas for the number and sum of divisors.

Problem 2.2. What's the sum of the square roots of the divisors of 2024?

Example 2.3 (PUMaC Number Theory A 2015/6)**.** Let $d(n)$ be the number of positive divisors of n. Compute the smallest n such that 35 divides

$$\sum_{t|n} d(t)^3.$$

Let's consider the function $f(n) = \sum_{t|n} d(t)^3$. We note that $d = \mathbf{1} * \mathbf{1}$ is multiplicative, so d^3 is multiplicative, so f is multiplicative. Let's try to compute f for prime powers p^α. We see

$$f(p^\alpha) = d(1)^3 + d(p)^3 + \cdots + d(p^\alpha)^3 = 1^3 + 2^3 + \cdots + (\alpha + 1)^3 = \left(\frac{(\alpha + 1)(\alpha + 2)}{2}\right)^2.$$

Therefore, writing n as its prime factorization gives

$$f(n) = f(p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}) = f(p_1^{\alpha_1}) f(p_2^{\alpha_2}) \cdots f(p_k^{\alpha_k}) = \prod_{p^\alpha \| n} \left(\frac{(\alpha + 1)(\alpha + 2)}{2}\right)^2.$$

We want the minimum n such that 35 divides this product, which is achieved when one of the α_i is 3 and another is 5, so the minimum possible n is $2^5 \cdot 3^3 = 864$.

3 Euler's totient function

Definition 3.1. Euler's totient function, denoted φ, returns the number of positive integers less than a given positive integer relatively prime to it.

Theorem 3.2. φ is multiplicative.

Proof. By the Chinese Remainder Theorem, we have

$$\varphi(mn) = \left|(\mathbb{Z}/mn\mathbb{Z})^\times\right| = \left|(\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times\right| = \varphi(m)\varphi(n).$$

□

Problem 3.3. Compute $\varphi(2024)$.

Theorem 3.4. For all n, $\sum_{d|n} \varphi(d) = n$, or $\varphi * \mathbf{1} = \mathrm{id}$.

We present two proofs of this fact.

Proof with prime powers. Since both sides are multiplicative, it's enough to show the equality for $n = 1$, which is clear, and prime powers $n = p^\alpha$. We have

$$\sum_{d|p^\alpha} \varphi(d) = \varphi(1) + \varphi(p) + \cdots + \varphi(p^n) = 1 + (p - 1) + \cdots + \left(p^\alpha - p^{\alpha-1}\right) = p^\alpha,$$

as desired. □

Proof by double-counting. Consider the fractions $\frac{1}{n}, \frac{2}{n}, \ldots, \frac{n}{n}$. We count the number of fractions by casework on their denominator when written in simplest form. If $d \mid n$, observe that there are $\varphi(\frac{n}{d})$ solutions $0 \le k \le n-1$ to $\gcd(k,n) = d$. Therefore, there are $\varphi\left(\frac{n}{\frac{n}{d}}\right) = \varphi(d)$ fractions with denominator d, so there are $\sum_{d|n} \varphi(d)$ fractions in total. But clearly there are n fractions, so $\sum_{d|n} \varphi(d) = n$, as desired. □

Problem 3.5 (CMIMC Algebra 2023/7)**.** Compute

$$\sum_{k=1}^{\varphi(2023)} \frac{\gcd\left(k, \varphi(2023)\right)}{\varphi(2023)}.$$

Problem 3.6 (PUMaC Number Theory A 2016/7)**.** Compute the number of integers n between 2017 and 2017^2 such that $n^n \equiv 1 \pmod{2017}$.

4 More theory

Definition 4.1. Let ε be the arithmetic function satisfying $\varepsilon(1) = 1$ and $\varepsilon(n) = 0$ for $n \ge 2$.

Note that ε is multiplicative. Also note ε is the identity under the Dirichlet convolution, since $\varepsilon * f = f$ for any arithmetic function f.[1]

Definition 4.2. Call a positive integer *squarefree* if no perfect squares divide it. Define the Möbius function

$$\mu(n) = \begin{cases} (-1)^k & \text{if } n \text{ is squarefree and is the product of } k \text{ distinct primes} \\ 0 & \text{if } n \text{ is not squarefree.} \end{cases}$$

Problem 4.3. Convince yourself μ is multiplicative.

Why is μ useful? Well, it turns out that it has a very special property:

Theorem 4.4. $\mu * \mathbf{1} = \varepsilon$.

Proof. We wish to show

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & n = 1 \\ 0 & \text{otherwise} \end{cases}$$

for all positive integers n. Since both sides are multiplicative, it's enough to show the equality for $n = 1$, which is clear, and prime powers $n = p^\alpha$. We see

$$\sum_{d|p^\alpha} \mu(d) = \mu(1) + \mu(p) + \mu(p^2) + \cdots + \mu(p^\alpha) = 1 + (-1) + 0 + \cdots + 0 = 0 = \varepsilon(p^\alpha),$$

as desired. □

This allows us to turn a summation function back into the original function.

Theorem 4.5 (Möbius inversion)**.** Let f be an arithmetic function and let F be the summation function of f. Then

$$F * \mu = f.$$

The proof is really easy.

[1] Don't confuse ε with id: ε is the identity under the Dirichlet convolution, and id is the identity function.

Proof.

$$F * \mu = f * \mathbf{1} * \mu = f * \varepsilon = f. \qquad \square$$

Just like $\mathbf{1}$ can be inverted, all arithmetic functions with $f(1) \neq 0$ can be inverted.

Theorem 4.6 (Dirichlet inverses)**.** If f is an arithmetic function with $f(1) \neq 0$, there exists a function g also with $g(1) \neq 0$ such that $f * g = \varepsilon$. Furthermore, g is multiplicative iff f is.

Proof. We claim the function g given by the recursive formula

$$g(n) = \begin{cases} \frac{1}{f(1)} & n = 1 \\ -\frac{1}{f(1)} \sum_{d|n, d<n} f(n) g\left(\frac{n}{d}\right) & n > 1 \end{cases}$$

works. We show $(f * g)(n) = \varepsilon(n)$ for all n. If $n = 1$, we see

$$(f * g)(1) = f(1)g(1) = f(1) \cdot \frac{1}{f(1)} = 1 = \varepsilon(1).$$

Now if $n > 1$, the definition of $g(n)$ becomes

$$g(n) = -\frac{1}{f(1)} \left((f * g)(n) - f(1)g(n) \right),$$

which rearranges to $(f * g)(n) = 0 = \varepsilon(n)$, so $f * g = \varepsilon$, as desired. $\square$

The set of arithmetic functions forms the *Dirichlet ring* with pointwise addition and multiplication given by the Dirichlet convolution.

5 Problems

Problem 5.1 (Berkeley Math Circle)**.** Suppose that we are given infinitely many tickets, each with one natural number on it. For any $n \in \mathbb{N}$, the number of tickets on which divisors of n are written is exactly n. For example, the divisors of 6, $\{1, 2, 3, 6\}$, are written in some variation on 6 tickets, and no other ticket has these numbers written on it. Prove that any number $n \in \mathbb{N}$ is written on at least one ticket.

Problem 5.2 (AMC 12A 2023/22)**.** Let f be an arithmetic function with $\mathrm{id} * f = \mathbf{1}$. Compute $f(2023)$.

Problem 5.3 (Brilliant)**.** Show that

$$\sum_{d|n} \sigma\left(\frac{n}{d}\right) \varphi(n) = nd(n).$$

Problem 5.4 (SMT Team 2024/14)**.** Let $f(n)$ be the number of positive divisors of n that are of the form $4k + 1$, for some integer k. Find the number of positive divisors of the sum of $f(k)$ across all positive divisors of $2^8 \cdot 29^{59} \cdot 59^{79} \cdot 79^{29}$.

Problem 5.5 (CMIMC Algebra 2020/8)**.** Let $f : \mathbb{N} \to (0, \infty)$ satisfy $\prod_{d|n} f(d) = 1$ for every n which is not prime. Determine the maximum possible number of n with $1 \leq n \leq 100$ and $f(n) \neq 1$.

6 Solutions

Solution to Problem 2.2. Define $\mathrm{id}_{1/2}(n) := \sqrt{n}$, and define $\sigma_{1/2} := \mathrm{id}_{1/2} * \mathbf{1}$. We wish to compute $\sigma_{1/2}(2024)$. Since $\mathrm{id}_{1/2}$ and $\mathbf{1}$ are multiplicative, so is $\sigma_{1/2}$. Therefore

$$\begin{aligned}\sigma_{1/2}(2024) &= \sigma_{1/2}(2^3)\sigma_{1/2}(11)\sigma_{1/2}(23)\\ &= (1+\sqrt{2}+\sqrt{4}+\sqrt{8})(1+\sqrt{11})(1+\sqrt{23})\\ &= 3(1+\sqrt{2})(1+\sqrt{11})(1+\sqrt{23}).\end{aligned}$$

■

Solution to Problem 3.3. Since φ is multiplicative, we only need to compute φ for prime powers. For a prime power p^α with $\alpha \neq 0$, $\varphi(p^\alpha) = p^\alpha - p^{\alpha-1}$, since a positive integer at most p is coprime with p^α iff it is not a multiple of p. Therefore, we have

$$\varphi(2024) = \varphi\left(2^3\right)\varphi(11)\varphi(23) = \left(2^3 - 2^2\right)(11-1)(23-1) = 880.$$

■

Solution to Problem 3.5. Let

$$f(n) = \frac{1}{n}\sum_{k=1}^{n} \gcd(n,k).$$

We want to find $f(\varphi(2023))$. If $d \mid n$, observe that there are $\varphi(\frac{n}{d})$ solutions $1 \le k \le n$ to $\gcd(k,n) = d$. Therefore, by casework on $\gcd(n,k)$,

$$f(n) = \frac{1}{n}\sum_{d\mid n} d \cdot \frac{n}{d} = \sum_{d\mid n} 1 = d(n).$$

Observing $\varphi(2023) = \varphi(7\cdot 17^2) = \varphi(7)\varphi(17^2) = (7-1)(17^2-17) = 2^5\cdot 3\cdot 17$, we see $f(\varphi(2023)) = d(\varphi(2023)) = (5+1)(1+1)(1+1) = 24$. ■

Solution to Problem 3.6. Note $n = 2017^2$ does not work. Observe the set of positive integers between 2017 and $2017^2 - 1$ are in bijection with the set of equivalence classes of ordered pairs $(a \mod 2017, b \mod 2016)$. Therefore, it's equivalent to find the number of ordered pairs (a,b) with $0 \le a \le 2016$ and $0 \le b \le 2015$ with $a^b \equiv 0 \pmod{2017}$.

Observe 2017 is prime, so for all $d \mid 2016$, there are $\varphi(d)$ a with order d, and if a has order d, there are $\frac{2016}{d}$ choices for b.

$$\sum_{d\mid 2016} \varphi(d)\frac{2016}{d}.$$

Let

$$f(n) = \sum_{d\mid n} \varphi(d)\frac{n}{d} = (\varphi * \mathrm{id})(n).$$

We see f is multiplicative, therefore,

$$\begin{aligned}f(2016) &= f(2^5\cdot 3^2\cdot 7)\\ &= f(2^5)f(3^2)f(7)\\ &= \left(2^5 + 5(2^5-2^4)\right)\left(3^2 + 2(3^2-3)\right)(7+7-1)\\ &= 30576.\end{aligned}$$

■

Solution to Problem 5.1. Let $f(n)$ denote the number of tickets n is written on. The condition translates to

$$\sum_{d|n} f(d) = n,$$

so $f(n) = \varphi(n)$, which is clearly always positive. ■

Solution to Problem 5.2. Since id and $\mathbf{1}$ are multiplicative, so is f. Now

$$f(2023) = f(7 \cdot 17^2) = f(7)f(17^2).$$

From plugging in $n = 1, 7, 17, 17^2$ into $(\mathrm{id} * f)(n) = \mathbf{1}(n)$, we see

$$\begin{aligned} 1f(1) &= 1 \\ 7f(1) + 1f(7) &= 1 \\ 17f(1) + 1f(17) &= 1 \\ 17^2 f(1) + 17f(17) + 1f(17^2) &= 1. \end{aligned}$$

Solving the system, we see $f(1) = 1$, $f(7) = -6$, $f(17) = -16$, and $f(17^2) = -16$. Therefore,

$$f(2023) = -6 \cdot -16 = 96.$$ ■

Solution to Problem 5.3. The left side is $(\sigma * \varphi)(n)$. From the Möbius inversion of $\varphi * \mathbf{1} = \mathrm{id}$, we see $\varphi = \mu * \mathrm{id}$. Therefore,

$$\sigma * \varphi = \mathrm{id} * \mathbf{1} * \mu * \mathrm{id} = \mathrm{id} * \mathrm{id}\,.$$

But

$$(\mathrm{id} * \mathrm{id})(n) = \sum_{d|n} d \cdot \frac{n}{d} = \sum_{d|n} n = nd(n).$$ ■

Solution to Problem 5.4. Let $A = 29^{59}$, $B = 59^{79}$, $C = 79^{29}$, and $N = ABC$. Let F be the summation function of f. Note that the powers of two "don't matter": $f(2^k n) = f(n)$, since we are only counting 1 (mod 4) divisors, which must be odd. Therefore, $F(2^k n) = (k+1)F(n)$ for odd integers n. We want to compute $F(2^8 N) = 9F(N)$.

We note f is not multiplicative: for example, $1 = f(3)f(7) \neq f(21) = 2$, so we cannot simply one-shot the problem by computing F for prime powers. Hmm...

To exploit symmetry, we are motivated to define $g(n)$ to return the number of 3 (mod 4) divisors of n, and G to be the summation function of g. Now since $1 \cdot 1 \equiv 3 \cdot 3 \equiv 1 \pmod 4$ and $1 \cdot 3 \equiv 3 \cdot 1 \equiv 3 \pmod 4$, for coprime m, n we have

$$\begin{aligned} f(mn) &= f(m)f(n) + g(m)g(n) \\ g(mn) &= f(m)g(n) + g(m)f(n). \end{aligned}$$

Now we compute

$$\begin{aligned} 9F(N) &= 9 \sum_{d|N} f(d) \\ &= 9 \sum_{a|A} \sum_{b|B} \sum_{c|C} f(abc) \\ &= 9 \sum_{a|A} \sum_{b|B} \sum_{c|C} f(a)f(b)f(c) + f(a)g(b)g(c) + g(a)f(b)g(c) + g(a)g(b)f(c) \end{aligned}$$

$$= 9(F(A)F(B)F(C) + F(A)G(B)G(C) + G(A)F(B)G(C) + G(A)G(B)G(C)).$$

But since $29 \equiv 1 \pmod 4$, A has no $3 \pmod 4$ divisors, so $g(a) = 0$ for all $a \mid A$, so $G(A) = 0$. Therefore, the expression becomes

$$9F(A)(F(B)F(C) + G(B)G(C)).$$

We compute $F(A) = T_{60}$, $F(B) = 2T_{40}$, $F(C) = 2T_{15}$, $G(B) = 2T_{40} - 40 = 40^2$, and $G(C) = 2T_{15} - 15 = 15^2$, where T_n denotes the nth triangular number. Since $T_n = \frac{n(n+1)}{2}$, the expression then becomes

$$9 \cdot \frac{60 \cdot 61}{2} \left(40 \cdot 41 \cdot 15 \cdot 16 + 40^2 \cdot 15^2\right) = 2^7 \cdot 3^4 \cdot 5^3 \cdot 61 \cdot 157,$$

which has 640 factors. ■

Solution to Problem 5.5. We've seen sums over the divisors of a number, but never products. This motivates us to turn the product into a sum by taking the logarithm of both sides and making the substitution $g(n) = \log(f(n))$. The condition becomes

$$\sum_{d|n} g(d) = 0$$

for non-prime n, and we'd like to maximize the number of nonzero values of n.

Although we do know something about the summation function of g, we don't know much about g itself. Therefore, we're motivated to use Möbius inversion. Define $G(n) = \sum_{g|n} g(d)$, so $G(n) = 0$ for all non-prime n. Now we see

$$g(n) = (G * \mu)(n) = \sum_{d|n} G(d)\mu\left(\frac{n}{d}\right).$$

Noting that $G(d)$ is zero if d is not prime, the sum becomes

$$\sum_{p|n} G(p)\mu\left(\frac{n}{p}\right).$$

Note that $g(1) = 0$. We disregard it in the following discussion.

Call an integer n *nice* if it is in the forms $p_1 \dots p_k$ or $p_1^2 p_2 \dots p_k$, and call one *mean* if it is not nice. To ensure $g(n)$ is nonzero, $\mu\left(\frac{n}{p}\right)$ must be nonzero for at least one p, so $\frac{n}{p}$ must be squarefree for at least one p, so n must be nice. Additionally, we may choose values of $g(2), g(3), g(5), \dots$ such that every nice n has $g(n)$ nonzero. For example, $g(p) = \sqrt{p}$ for primes p guarantees all nice n have $g(n)$ nonzero since all $\sqrt{p}$ are linearly independent over $\mathbb{Q}$. Therefore, the answer is the number of nice n at most 100.

We complementary count. We want the number of mean integers less than 100. Note that an integer is mean iff a perfect cube not equal to 1 or the square of a composite number divides it. Since we are only considering integers at most 100, we only need to check for divisibility by 8, 27, 36, and 100. The mean integers are the multiples of 8 between 8 and 96, 27, 54, 81, 36, and 100, for a total of 17 mean integers. Remembering to count 1, the number of integers satisfying the condition is 82. ■